FENG-AN (Alan) HSIEH

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Summary

Engineer with 4 years of C/C++ development at the hardware/firmware boundary: modeling hardware IP from RTL specifications, enabling firmware bring-up and debug on virtual platforms with processor fast models, and writing Python tooling and automation for firmware teams. Experienced in HW/SW co-simulation and in isolating bugs across HW and FW. Built AI-agent workflows that cut model delivery from ∼2 weeks to 2–3 days.

Experience

SiliconMotion – ESL Modeling Team **Zhubei, Hsinchu**
Chief Engineer *July 2026 – Present*

- Own 6 C++/SystemC TLM IP models written from RTL design specifications and delivered to 5+ firmware engineers, who use them to develop and debug firmware before silicon; models are also packaged into virtual platforms for external customers.
- Modeled controller data-path and control-path logic (command arbitration, DMA, error handling, command decode) and a shared inter-engine messaging IP that handshakes with every engine.
- Built an AI agent harness and reusable agent skills (Claude Code, Cursor) for spec-to-model scaffolding, documentation sync, and automated regression.
- Mentor engineers on TLM-2.0 modeling standards and development workflow.

Senior Engineer *November 2022 – June 2026*

- Built and maintained full-system virtual platforms on Synopsys VDK, integrating processor fast models (Arm Cortex-R82, Cortex-R8, Cortex-A55; Andes D23) with peripheral TLM models, so firmware teams could bring up and test software months ahead of silicon.
- Ran HW/SW co-simulation with DV patterns and verified models against RTL, finding and resolving CModel and firmware bugs at the HW/FW boundary.
- Built Python tools for platform-level trace analysis and debugging, shortening issue isolation for firmware teams.
- Automated RTL waveform (FSDB/VCD) parsing with Python and shell, speeding up test creation by 3x and saving ∼1 week of manual effort per cycle.
- Profiled and tuned processor fast models and platform integration to improve simulation speed and stability.

Skills

Embedded & Firmware-facing: C, C++, Assembly, Arm Cortex-R/A and Andes cores (fast models), Firmware Debug on Virtual Platforms, HW/SW Co-simulation, Timer/Peripheral IP Modeling
Scripting & Tooling: Python (test automation, trace analysis, waveform parsing), Shell, Git
Modeling & Verification: SystemC, TLM-2.0, Synopsys VDK, Verilog, RTL-vs-model Verification, FSDB/VCD
Protocols & Architecture: AMBA spec & bus integration (AXI, AHB, APB, CHI), DMA
Automation & AI: Claude Code, Cursor, Agent Skills, Regression Flows
Languages: Mandarin (Native), English (TOEIC 780)

Education

National Chung Cheng University **Chiayi, Taiwan**
M.E. in Electrical Engineering *August 2020 – July 2022*
Thesis: An End-To-End System For Road Agent Behavior Classification (Deep Learning, Computer Vision)
National University of Kaohsiung **Kaohsiung, Taiwan**
B.E. in Computer Science and Information Engineering *September 2016 – June 2020*